Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | A Hybrid Memory System for Continuous Learning in Multi-Agent Robotic Fleets | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Knowledge Services | | Mechnical/Materials | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *R&D / Engineering Services* | *Other R&D / Engineering-Related Technologies* | *Robotics/ Automation Machinery* | *Robotics/Automation Machinery Software* |
| Concept of Proposed Technology | Deploying AI agents in robotics is currently hindered by dynamic environments, that require frequent adaption of the AI agents. Regular retraining to learn new information about the environment is prohibitively expense. To address this adaptation bottleneck, we propose a Hierarchical Hybrid Memory Architecture that decouples immediate information retrieval from foundational model updates. The core of this system is a Short-Term memory module designed for resource-constrained edge hardware. This low-latency, low-capacity layer allows individual agents to instantly "cache" and retrieve context-specific data, such as temporary environmental changes or task-specific parameters, without requiring a full weight update to the neural network. This ensures real-time adaptability and operational continuity on the edge, mimicking human working memory. Complementing the edge layer, the architecture integrates a Long-Term memory system: a high-capacity, centralized repository that aggregates experiences from multiple agents to enable | | | |

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| | information exchange between multiple AI agents working in the same environment. This layer serves as a shared knowledge base, allowing the fleet to generalize learnings from individual encounters. Crucially, this dual-tier approach renders lifelong learning economically viable by facilitating periodic offline consolidation. Instead of continuous, expensive retraining, the system selectively utilizes the aggregated high-value data from Long term storage to retrain the core models during scheduled downtime. |
| **Rationale for Joint R&D Project** | This research project requires both expertise in the development of trustworthy AI systems as well as optimization and integration into edge devices. Germany can provide its extensive expertise in the development of cognitive architectures, resilient system architectures and the development of AI systems. Furthermore, Germany contributes expertise in data safety and privacy regulations that affect industrial robotics applications but especially applications in medical robotics.<br>South Korea brings expertise in hardware integration, advanced connectivity (5G/6G) and rapid commercial prototyping especially in electronics and robotics. |
| **Key Words** | 1. Edge AI<br>2. Continual Learning<br>3. Hierarchical Memory<br>4. Multi-Agent Robotics |

## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level (3) | End of the Project | Level (6) |
|---|---|---|---|---|
| **R&D Objectives** | This project aims to develop a system demonstrator the for Hierarchical Hybrid Memory A rchitecture. This involves transitioning from isolated software simulations to a robust, hard ware-accelerated prototype that functions reliably under real-world constraints. Specifically, the project aims to achieve the following technical milestones:<br>• **Develop a Latency-Critical Edge Memory:** Engineer a lightweight, short-term memory module optimized for edge-class hardware (e.g., NVIDIA Jetson or similar embedded NPUs), capable of short retrieval times for immediate context awareness without acces sing the cloud.<br>• **Establish a Scalable Long-Term Knowledge Base:** Construct a centralized, high-cap acity vector database that aggregates experience from multiple heterogeneous agents, enabling semantic search and cross-agent knowledge transfer with high fidelity.<br>• **Implement "Offline Consolidation" Protocols:** Develop and validate the selective forg etting and consolidation algorithms that govern the periodic transfer of high-value data f rom Short to Long term moery during charging/idle phases, ensuring model updates oc cur without catastrophic forgetting.<br>• **Validate in a Relevant Environment:** Demonstrate the full closed-loop system in a ph ysical testbed (e.g., a mock logistics or service robot scenario), proving that agents can autonomously adapt to new tasks and share learned behaviors over a continuous 48-hour operation cycle. | | | |
| **R&D Target Specifications** | • Achieve a read/write latency of < 100 ms for local context queries on standard embedded hardware (e.g., NVIDIA Jetson Orin), ensuring near real-time decision-making capabilities without cloud dependency.<br>• Demonstrate successful synchronization and information sharing between a minimum of 3 heterogeneous agents simultaneously, with a data consistency update rate of < 5 seconds to the shared long-term memory.<br>• Reduce the dataset size required for effective model fine-tuning by > 20% (compared to full historical retraining) by using selective memory archival, while maintaining the model's accuracy on previous tasks within 10% of the baseline (mitigating catastrophic forgetting).<br>• Achieve a continuous, autonomous runtime in the target test environment without memory leaks, system crashes, or degradation in retrieval speeds. | | | |
| **R&D Plan** | **Year 1: Requirements analysis & Independent Memory modules**<br>Goal: Develop the independent Tier 1 and Tier 2 modules before integration.<br>• WP1: Requirements & Architecture (Joint)<br>  ○ Define the data packet structure for "Memory Objects" (what gets saved).<br>  ○ Set strict latency budgets for Edge vs. Cloud.<br>• WP2: Tier 1 Edge Memory Module<br>  ○ Develop the NPU-accelerated vector store for the robot.<br>  ○ Optimizing for low-power edge devices.<br>• WP3: Tier 2 Shared Knowledge Base<br>  ○ Construct the central semantic database (Knowledge Graph + Vector DB).<br>  ○ Develop privacy-preserving retrieval algorithms for multi-agent access. | | | |

**Year 2: Integrate Long and Short term memory modules into joint system**

Goal: Connect the Edge and Cloud, enabling the "Offline Consolidation" cycle.

- WP4: Synchronization Protocol
  - Build the 5G/6G communication middleware.
  - Implement "Priority Queuing": Deciding which memories are urgent (upload now) vs. archival (upload during charging).
- WP5: Offline Consolidation & Retraining
  - Develop the "Sleep Phase" algorithm: The system analyzes Tier 2 data overnight to fine-tune the core model.
  - Mitigate "Catastrophic Forgetting" during these updates.
- WP6: Hardware Integration
  - Install the software stack onto physical robotic prototypes (e.g., mobile service robots or AMRs).

**Year 3: Validation & Pilot**

Goal: Prove economic viability and technical stability in a real scenario.

- WP7: Pilot Testing (Joint)
  - Deploy 3+ robots in a mock warehouse or service environment.
  - Test Scenario: Robot A learns a new obstacle; Robot B "downloads" this memory and avoids it without prior experience.
- WP8: Optimization & Benchmarking
  - Measure latency against the <100ms target.
  - Refine energy consumption of the memory module.

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | • Develop memory architecture and software: low latency Vector Databases, query s ystems etc.<br>• Implementing synchronization mechanism between long- and short-term memory<br>• Implementing offline information consolidation and retraining paragraphs ensuring c ontinuous learning of the AI agents<br>• Implement data safety and privacy mechanism to ensure compliance with regulatio ns |
| **Expected R&R of Korean SME Partner** | • Adaption and optimization of memory systems to edge devices on the robots<br>• Implementation of hardware accelerators systems (e.g. NPU, Edge GPU or similar) for edge AI agent<br>• Integration with AI agents in a robotic use case<br>• Provide robots and edge devices for demonstrator |

## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | By successfully decoupling immediate behavioral adaptation from expensive model retraining, the project will demonstrate a reduction in fleet maintenance costs, making advanced adaptive AI economically viable for SMEs and logistics operators who currently cannot afford continuous model updates. Beyond immediate economic gains, the project establishes a strategic technological bridge between Germany and South Korea: it combines German leadership in trustworthy cognitive architectures with South Korean dominance in 5G-connected hardware and semiconductor integration. This synergy will result in a jointly owned intellectual property portfolio for Memory-Augmented Robotics, positioning the consortium to define upcoming ISO standards for lifelong learning in autonomous systems and opening new export markets for both nations in the rapidly growing service robotics sector. The developed technologies can be commercialized through Korean robotics SMEs and scaled across multiple verticals e.g. industrial or medical robotics. |

## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
|---|---|---|
| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |